

History-dependent state selection after transient coupling loss in a conductance-based bioelectric network

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Abstract

Background: Bioelectric bistability and memory are established phenomena, but restoring intercellular coupling does not necessarily restore the previous electrical state. We examined whether a temporary coupling perturbation selects different stable states when the final weighted network and external input are identical.

Methods: A synthetic network of 226 cells and 622 undirected connections combined leak and voltage-dependent potassium currents with either a persistent or an inactivating depolarizing current. We compared uninterrupted coupling with temporary interface weakening followed by exact restoration. Local equilibria, triangular current ramps, network stationarity and Jacobian eigenvalues distinguished kinetic lag from persistent state selection. A frozen validation comprised 136 unique network conditions and six isolated-cell protocols, including source/geometry variants, a sampled parameter map and an alternative cubic bistable mechanism.

Results: At the lower fixed stimulus, restored networks recruited all receivers while intact networks retained localized activation in all five source/geometry configurations. History-dependent final-state differences occurred in 24 of 42 interface comparisons across the sampled conductance-restoration grid. The cubic mechanism had two stable equilibria and demonstrable hysteretic switching, yet all 21 stimulated network conditions returned to rest without receiver recruitment. All validation conditions passed refinement checks, with unchanged classifications in 44 independent-solver comparisons. Earlier numerical discrepancies and their corrections were retained.

Conclusion: In the tested synthetic conductance-based network, transient perturbation of intercellular coupling altered stable state selection even after exact restoration of the original weighted connectivity. This is a controlled extension of established attractor and bioelectric-memory concepts. The negative cubic comparison shows that bistability and hysteresis alone did not suffice under the fixed stimulation protocol; physiological generality and priority of the exact comparison remain unestablished.

Introduction

Membrane potential, V_{mem} , depends on membrane currents and electrical exchange with neighboring cells. Gap junctions therefore make the state of a cell depend on both its own membrane dynamics and the surrounding tissue. This coupling can support spatial voltage patterns and collective responses. However, neither the existence of voltage-dependent channels nor a visible loop during a parameter sweep establishes regenerative excitability or persistent memory.

Local bistability requires coexistence of distinct stable states at the same parameter values. In a bistable system, a transient intervention can change which state is reached after the intervention ends. History dependence is consequently a familiar property of nonlinear dynamics. Associative attractor networks and basin-directed network control provide general precedents; these ideas do not originate with bioelectric models. [Hopfield, 1982](#); [Cornelius et al., 2013](#).

Bioelectric studies likewise precede this work. Single-cell conductance models demonstrated switchable resting-potential memories, and multicellular models established coupling-dependent voltage patterns. [Law and Levin, 2015](#); [Cervera et al., 2016](#). BGR models connected voltage feedback, hysteresis and junction-dependent patterning. Engineered tissues subsequently provided experimental evidence for electrical bistability and collective domain transitions. [Pietak and Levin, 2017](#); [McNamara et al., 2020](#).

Can transient perturbation of intercellular coupling alter stable electrical state selection even after exact restoration of the original weighted network connectivity? This comparison is experimentally difficult but computationally identifiable. Drug removal in an organism does not by itself certify equality of every junctional conductance. Conversely, a difference in final voltage while junctions remain weakened cannot isolate a memory of previous coupling from the effect of continuing damage.

Our aim was to make that distinction operational. We used a reduced synthetic conductance model, imposed a finite interface perturbation, restored every affected weight exactly and assessed the resulting equilibria. We also tested whether the outcome persisted under a bounded set of source and geometry changes, and whether an independently specified cubic bistable membrane model produced the same network response. These tests address mechanism and protocol dependence; they do not establish a new general law of memory or a physiological mechanism of regeneration.

Methods

Study organization and frozen protocols

The study comprised a developmental/main campaign followed by a directed final validation. The main campaign contained 487 labeled conditions, including temporal-transmission, memory, conductance-sensitivity and partial-repair conditions and controls. Labels are not independent biological replicates, and some controls were repeated across groups. Parameter exploration on isolated cells and the original network preceded the main protocol. Additional current-ablation controls, equilibrium checks and numerical corrections were explicitly recorded as subsequent analyses.

The final validation contained 136 unique network conditions and six isolated-cell protocols. Its three blocks had 65, 49 and 24 memberships, with two shared network conditions: 138 memberships are therefore not 138 independent network cases. The protocol was frozen locally before those outcomes were generated. Prior knowledge from the original network informed the design; this was prospective protocol execution conditional on earlier exploration, not blinded

validation of an independently derived biological model. No new simulations were performed during manuscript preparation.

The main protocol SHA-256 is 5cbb0be57d3d0a06bf0f8ff971c9db2ded15500a959596ed44f1eb7db40680ba; the final-validation protocol SHA-256 is a9d5d8078bfef94c02864a6c7753946882259e0846e44b8541caddb715deac4fb. The local freeze times are recorded as UTC in the archived protocols. The publication package distinguishes these original hashes from any later metadata-only redaction in a public copy.

Network and membrane equations

The original graph contained 226 nodes and 622 undirected edges, with unit baseline weights. Geometry and adjacency were reused from an archived computational tissue configuration. The present experiments used an independent reduced ODE implementation. They were not new runs of native BETSE tissue dynamics, and the weighted perturbed graphs were not verified as native BETSE FAST executions.

Let W be the symmetric weighted adjacency, $L(W)$ its unnormalized Laplacian and G_j the coupling scale. For node i ,

$$C_m \dot{V}_i = -G_j[L(W)V]_i - G_L(V_i - E_L) - G_K m_i(V_i - E_K) + g_D a_\infty(V_i) h_i(E_D - V_i) + I_i(t)$$

$$\dot{m}_i = [m_\infty(V_i) - m_i]/\tau_K, \quad m_\infty(V) = \text{logistic}[(V + 0.00076)/0.01409]$$

$$a_\infty(V) = \text{logistic}[(V + 0.025)/0.004].$$

Voltages are in volts and time in seconds. Parameters were $C_m = 0.01 \text{ F m}^{-2}$, $G_L = 1 \text{ S m}^{-2}$, $G_K = 2 \text{ S m}^{-2}$, $E_L = -0.05 \text{ V}$, $E_K = -0.0796750718514 \text{ V}$ and $E_D = +0.05 \text{ V}$. Conductance densities and stimulus current density are effective model quantities under a common normalization, not measurements of individual gap-junction conductance in siemens. Baseline G_j was 1 S m^{-2} in the main persistent-state comparison and validation.

The potassium activation midpoint and slope originate from published heterologous hKv1.5 measurements. The constant $\tau_K = 14.8 \text{ ms}$ is an approximation based on one reported deactivation timescale, not an identified $\tau(V)$. The remaining membrane parameters and the depolarizing mechanism are synthetic. This combination is not a calibrated cell model. [Dong et al., 2022](#).

The graded comparator used $g_D = 0$. The persistent variant used $h = 1$ and, nominally, $g_D = 4 \text{ S m}^{-2}$. For the excitable variant, h obeyed

$$\dot{h}_i = [h_\infty(V_i) - h_i]/0.1, \quad h_\infty(V) = \text{logistic}[-(V + 0.040)/0.004].$$

The added depolarizing term supplied regenerative feedback. Voltage dependence of the potassium term alone was not treated as evidence for regenerative firing. The persistent and inactivating variants were analyzed separately because stable recruitment and transient excitation are different endpoints.

Stimulation, interface perturbation and exact restoration

For the original network, stimulation targeted node 61, with 104 receivers on the designated side of the source. The interface consisted of 32 crossing edges. Weakening each edge from 1 to 0.1 removed 28.8 weight units while preserving a nonzero path. Matched random controls weakened the same number of edges by the same fraction. Complete interface cutting, used in additional controls, was distinguished from 90% weakening.

The persistent-network stimulus was a 20 ms current pulse. Its reference upper amplitude, $0.082373046875 \text{ A m}^{-2}$, was determined during earlier original-network development by a threshold search; the lower bracketing amplitude was $0.08232421875 \text{ A m}^{-2}$. The final validation used factors 0.9, 1.0 and 1.1 of the upper reference without recalibration for geometry or the cubic mechanism. Thus factor 0.9 corresponds to $0.0741357421875 \text{ A m}^{-2}$. A separate excitable reference was $0.12783203125 \text{ A m}^{-2}$. These are protocol anchors, not universal tissue thresholds.

In the central comparison, weakening began at the start of stimulation and the full original adjacency was restored at the scheduled time, without resetting any state variable. Initial states and applied current were otherwise matched. Integration was segmented at pulse and restoration discontinuities. After restoration and pulse termination, matched conditions shared the complete final vector field, not merely the number of edges. Stimulus-free, regenerative-current ablation and never-restored controls tested alternative interpretations.

Temporal graded-transmission experiments included fixed charge density, 0.0001 C m^{-2} , delivered over 1, 10 or 100 ms, and a fixed-current series at 0.02 A m^{-2} for the same durations. These experiments measured amplitude and integrated response and included changes in potassium gating timescale and G_j. Their normalized peak measures are not interchangeable with the later binary recruitment endpoint.

Hysteresis, equilibria and stability

We inspected triangular current ramps with 0.2, 1 and 5 s legs, held states at zero current and solved the steady membrane-current equation. A nonzero loop area was regarded as a descriptor of finite-rate trajectories, not a sufficient memory test. For the persistent mechanism, the ramp covered the two computed folds; the resulting current and voltage range was deliberately synthetic and extended beyond plausible physiological voltages.

At $gD = 4 \text{ S m}^{-2}$ and zero current, the persistent model had equilibria at -50.97123 , -36.01657 and $+7.54715 \text{ mV}$. The outer equilibria were stable and the intermediate equilibrium unstable in the isolated-cell Jacobian. We additionally assessed selected full-network equilibria, not only isolated-cell slopes. Small spatial perturbations of selected final network states and extension from 2 to 5 s were supporting checks. These do not map global basins or prove all possible long-time behaviors.

Directed final validation

Block 1 comprised the original configuration, left and right source/receiver variants, and two geometry variants. All graphs retained 226 nodes and 622 edges. The geometries used Gaussian positional jitter with standard deviation 0.1 times the original median edge length, fixed seeds 1701 and 2701, and a specified connectivity reconstruction. Only six and five edge replacements, respectively, distinguished the resulting geometries from the original. These are correlated, small perturbations, not samples from unrelated tissue architectures.

Receiver counts across the five configurations were 104, 168, 170, 110 and 109; interface sizes were 32, 28, 28, 37 and 38 edges. Source selection and receiver masks were defined before validation outcomes. Block 1 compared original, interface and one matched random weakening per configuration at the three amplitude factors, with restoration at 500 ms, plus no-stimulus, ablation and never-restored controls.

Block 2 retained the original geometry and factor 0.9. It sampled $gD = 3.4, 3.6, 3.8, 4.0, 4.2, 4.4$ and 4.6 S m^{-2} , with restoration at 20, 50, 100, 200, 350 and 500 ms. Seven intact references were compared with 42 interface conditions. We did not interpolate this grid into a proven continuous phase boundary.

Alternative cubic membrane mechanism

Block 3 replaced the entire leak-potassium-regenerative membrane current by a scalar cubic drift,

$$\dot{V}_i = \frac{(V_i - V_L)(V_i - V_U)(V_H - V_i)}{\tau_c(V_H - V_L)^2} - \frac{G_j}{C_m}[L(W)V]_i + \frac{I_i(t)}{C_m}$$

with $V_L = -0.05$ V, $V_U = -0.035$ V, $V_H = +0.01$ V and $\tau_c = 0.003125$ s. This is a phenomenological net membrane-current formulation with a different state dimension. It does not retain an additional potassium gate. Its low-state slope, -80 s⁻¹, was selected before network testing to approximate the earlier persistent model's low-state relaxation rate; high-state regenerative drive and network stimulation thresholds were not matched.

The cubic roots, local eigenvalues, four perturbed holds and two current ramps with 1 and 5 s legs were checked before network results. Network testing used the fixed persistent-model pulse amplitudes without retuning. The block contained 21 stimulated conditions, one no-stimulus condition and two leak-only ablations. Its negative outcome was retained under the prespecified stopping rule.

Endpoint definitions and numerical checks

In final validation, stationarity required $\max |dV/dt| < 10^{-5}$ V s⁻¹ and, where applicable, maximum gate derivative $< 10^{-4}$ s⁻¹. Stability required the largest real part of the full-network Jacobian eigenvalues to be below -10^{-6} s⁻¹. Departure from rest was assessed using a 0.1 mV tolerance. Recruitment used the intermediate zero-current equilibrium of the relevant model as threshold; at least half the receivers had to lie above it at the accepted final state for the collective category. This operational threshold is not an action-potential criterion.

History-dependent state selection required two accepted stable final states, exactly equal final weighted adjacency matrices and a maximum final voltage difference exceeding 1 mV. A source crossing followed by return to rest was classified as transient without persistence. Crossing the intermediate root alone was not called regenerative firing. For ablations lacking three roots, archived classification conventions were retained and explicitly separated from physiological spike interpretation.

All validation trajectories extended to 5 s and were sampled at 1 ms intervals. Primary DOP853 integration used $\text{rtol } 10^{-10}$, $\text{atol } 10^{-12}$ and maximum step 2.5 ms. Refinement used $\text{rtol } 10^{-12}$, $\text{atol } 10^{-14}$ and maximum step 1 ms. Forty-four prespecified network conditions also used Radau with $\text{rtol } 10^{-11}$, $\text{atol } 10^{-13}$ and maximum step 2.5 ms. The six isolated-cell protocols each had corresponding numerical checks. Numerical trajectories are technical verification runs, not additional experimental units. Software versions were Python 3.9.6, NumPy 1.26.4, SciPy 1.13.1 and Matplotlib 3.7.5.

No biological sampling model, inferential p-value or confidence interval over tissues was applied. Reported proportions enumerate the fixed computational conditions. Hash checks and software tests establish computational provenance and consistency, not physiological validity.

Partial repair as a separate endpoint

Earlier repair experiments restored only 4, 8 or 16 of the 32 weakened interface edges before stimulation. A directed ranking used the time integral of $|V_i - V_j|$ in a separate original-network graded response; three random repair orders were comparators. This was a specified heuristic ranking, not a proven optimum. It differs from the complete post-perturbation restoration protocol used for the central history comparison.

Results

Graded transmission remained sensitive to interface placement

In the main fixed-charge temporal experiments, the weakened-interface peak response retained approximately 18.00–38.07% of its matched original-network value across pulse duration and parameter combinations. Earlier matched-loss experiments used a different protocol and are not pooled with these measurements. Interface location therefore mattered for graded transmission in these networks, but the measured proportion was protocol dependent.

This result provides a reference for subsequent nonlinear behavior. It is not a direct measurement of a maximum-flow quantity or a Cheeger bound. Graph-cut and spectral arguments motivate attention to bottlenecks but do not determine the fraction transmitted by this dynamical membrane model.

Finite-rate loops distinguished kinetic lag from persistent bistability

The graded potassium model's loop area decreased from approximately 0.003282 to 0.000679 and 0.000136 V A m⁻² as ramp-leg duration increased from 0.2 to 1 and 5 s. Both tested zero-current starting states approached the same resting equilibrium. Together, these observations support kinetic lag in the sampled protocol (Figure 1).

The persistent variant retained two stable zero-current states. The 1 and 5 s ramps switched between branches and had areas of approximately 0.02472 and 0.02385 V A m⁻². The 0.2 s ramp did not provide the same switching result. Thus persistence was established by coexistence and stability, with ramp behavior as supporting evidence; neither a particular loop area nor a fully established quasistatic limit was required.

Local and network equilibrium checks supported persistent endpoints

For the persistent isolated model, largest real eigenvalues at the low and high equilibria were approximately -79.22 and -104.27 s⁻¹; the intermediate equilibrium had a positive eigenvalue. In the selected intact and restored network witnesses, corresponding full-state stability values were approximately -59.82 and -80.91 s⁻¹. Spatial perturbations of ±0.1 mV relaxed back in the tested checks, and extending the selected trajectories to 5 s did not change their classification.

The inactivating variant exhibited a distinct excitable regime. In isolated-cell checks, an above-threshold short pulse was followed by a regenerative voltage excursion and recovery, with recovery-dependent responsiveness to a second pulse. In network experiments, early restoration could permit a propagating transient while later restoration did not; the tested final states returned to rest. These outcomes are not evidence of the persistent network memory reported below.

The same final weighted graph supported different stable electrical states

At factor 0.9 in the original configuration, the intact network retained one high-state cell and no high-state receivers. With temporary interface weakening and restoration at 500 ms, all 226 cells, including all 104 receivers, reached the high state near +7.547 mV. Networks with identical final weighted adjacency, identical membrane equations and zero final external input reached distinct stable electrical states depending on their preceding coupling history. Figure 2 displays the two histories and endpoints.

Complete cutting followed by restoration also admitted later recruitment in the additional main-campaign control, whereas the corresponding never-restored cut did not transmit to the receivers. This distinguishes state selection after reconnection from an assertion that a signal physically crosses a missing connection. The 10%-weight interface is a weak connection and must not be described as an absent path.

The lower-amplitude contrast persisted across the five specified configurations

All five configurations reproduced the intact-localized versus restored-collective contrast at factor 0.9 (Figure 3). At factors 1.0 and 1.1, intact and restored networks both recruited collectively. Consequently, five of the 15 amplitude-by-configuration comparisons showed distinct final states. The result persisted across the selected source/mask changes and small geometric variations but depended on stimulation level.

The random-weakening control also recruited collectively at factor 0.9 in four configurations; the right-source configuration remained localized. History-dependent recruitment was therefore not unique to the named interface. Furthermore, one geometry variant recruited even without restoration of the weakened interface. Restoration is essential to making the final-structure comparison identifiable, but was not necessary for propagation in every sampled condition.

The sampled parameter map restricted the history-dependent region

All 42 weakened-and-restored interface conditions recruited collectively (Figure 4). Their intact references differed at $gD = 3.4\text{--}4.0 \text{ S m}^{-2}$, yielding history-dependent final-state differences in 24 of 42 comparisons. At 3.4 and 3.6, intact responses returned to rest after a transient; at 3.8 and 4.0, they remained localized. At $gD = 4.2\text{--}4.6$, intact references were already collective and final-state differences disappeared.

The persistent-model contrast occurred at every tested restoration time, 20–500 ms, within the lower four conductance values. The sampled map therefore does not support a narrow restoration-time window in that range. It supports a stimulus-mechanism-dependent contrast whose boundary was not located continuously.

Verified cubic bistability did not yield history-dependent network recruitment

Figure 5 compares the two current laws and the fixed-stimulus network endpoints.

The isolated cubic mechanism had three zero-current roots with local eigenvalues -80 , $+60$ and -240 s^{-1} . Perturbed holds returned to the two stable roots, and both tested ramp speeds produced switching in both directions. Its local bistability and hysteresis were therefore present before the network comparison.

Nevertheless, all 21 stimulated cubic network conditions returned to the low state, and no receiver crossed into the high state. The source did cross the isolated unstable voltage in all 21 conditions, illustrating that crossing that voltage at one node is insufficient to certify network basin entry. Twelve matched restored-versus-intact comparisons had no final-state difference beyond numerical rounding, with a maximum difference of approximately $6.94 \times 10^{-18} \text{ V}$.

This is a negative transfer result under fixed stimulation, not a failed bistable model. Matching approximate low-state relaxation and voltage scale did not match the nonlinear current shape, collective threshold or basin accessibility. The data do not distinguish which of those differences accounts for the outcome.

A graded-response repair ranking did not preserve the collective-response ordering

The directed partial-repair heuristic increased the graded receiver peak relative to the sampled random repairs in all 27 comparisons, by approximately 0.00810–0.04001 mV. In the persistent variant at factor 0.9, the directed repairs left zero of 104 receivers high, whereas the corresponding random repairs recruited all 104. At higher tested amplitudes, both strategies were collective.

Thus a repair ranking favorable for a graded peak was not favorable for the thresholded collective endpoint in this sampled setting. This secondary result concerns partial repairs before stimulation. It neither proves a globally optimal repair policy nor demonstrates that all linear or graded objectives invert in nonlinear systems.

Numerical checks retained failures and their resolution

In the main campaign, six initial refinement comparisons exceeded the 0.02 mV criterion; the largest discrepancy was approximately 0.03176 mV. The original trajectories and failed checks were retained. Finer replacements reduced the largest discrepancy to approximately 0.0002102 mV, without changing the scientific classification. Subsequent independent-solver checks supported the corrected trajectories.

All 136 final-validation network conditions passed refinement without replacement; the largest difference was approximately 0.0002149 mV. All 44 Radau comparisons retained classifications, with maximum voltage discrepancy approximately 1.115×10^{-11} V. Every accepted final-validation state satisfied the specified stationarity and full-Jacobian criteria. The final audit rechecked 87 paired comparisons and verified preservation of 584 main-campaign files. The editorial freeze subsequently verified both campaigns together, totaling 959 files including their manifests.

Discussion

Specific computational contribution

The defensible contribution is a controlled extension of coupling-dependent bioelectric state selection: two histories are compared under identical final weighted connectivity, stable electrical endpoints are numerically checked, and transfer to a second locally bistable mechanism is explicitly tested. We do not claim the first demonstration of bioelectric memory or of the general possibility that restored parameters can coexist with different states.

A targeted, non-exhaustive literature assessment did not establish priority of the exact conjunction of controls. Several relevant papers were accessible only at abstract level. In particular, the head-tail model of Cervera and colleagues is close enough that it prevents an unqualified first-report claim. [Cervera et al., 2020](#).

What was already known

Pezzulo, LaPalme, Durant and Levin explicitly reviewed bistability of somatic pattern memories and proposed an interpretation of stochastic planarian regeneration in terms of bioelectric memory. Their discussion includes persistent pattern changes following brief perturbations of intercellular electrical communication. This is a direct conceptual precedent for the present work. Our narrower contribution concerns paired electrical endpoints after exact restoration of the weighted network, numerical stability checks and a fixed-protocol comparison with a second bistable mechanism; it does not establish anatomical pattern memory or priority for bioelectric memory. [Pezzulo et al., 2021](#).

Temporary gap-junction blockade can have persistent biological consequences, although drug washout does not measure equality of all final junctional weights. [Durant et al., 2017](#). Mathematical network hysteresis likewise shows that restoring an interaction parameter need not restore a previous state. [Jiang et al., 2019](#). Partial uncoupling can also improve propagation through source-sink effects, so an apparently paradoxical benefit of weakening is not generally new. [Rohr et al., 1997](#).

The recent hybrid framework of Cortés-Poza includes bistability, adaptive gap junctions, attractor switching and additional regulatory variables. Its examined switching experiments apply transient external currents while conductances continue to adapt; they do not impose restoration of every original weight. This distinction identifies a different control variable, not a new mathematical principle. [Cortés-Poza, 2026](#).

From hysteresis to history-dependent state selection

The sequence “local multistability → possible path dependence → multiple network basins → trajectory-dependent final-state selection” is an interpretive framework subject to conditions. Local bistability permits different resting states; it does not specify which spatial states remain stable when cells are coupled. Nor does it specify whether a particular input and coupling history can reach another basin.

The ramp results first exclude a simpler explanation based solely on a finite-rate potassium loop. The equilibrium and network checks then establish persistent endpoints. Temporarily changing W changes the trajectory before both conditions again obey the same vector field. Their subsequent convergence to distinct stable states is consistent with entry into different attraction basins. We did not map the full basin boundary or isolate a single causal factor responsible for its crossing. Local multistability and hysteresis provide a dynamical framework for history dependence, but do not by themselves guarantee network-level state selection.

Why identical final connectivity matters

Exact restoration of the weighted adjacency removes persistent structural damage within the model as an explanation for the final electrical difference. Equal edge counts or equal unweighted support would be weaker controls. With the same membrane equations, zero input and exactly restored weights, the remaining difference resides in state, including gate variables, selected by the preceding dynamics. No additional explicit long-lived biochemical memory variable was required in this implementation.

This result does not imply storage of anatomical targets. It concerns stable electrical states in a fixed population with no cell division, differentiation, migration or tissue reconstruction.

Why bistability and hysteresis alone were insufficient

The cubic comparison is central because its negative network result cannot be attributed to absence of local bistability. Stable low and high roots and ramp switching were independently verified. A coupled network nevertheless imposes spatial electrical loading and a collective recruitment requirement that are absent from the isolated-cell assay.

Bistability and hysteresis alone were insufficient to produce network-level history dependence under the fixed stimulation protocol. This does not establish a universal distinction between cubic and conductance-based systems. Current shape, regenerative strength, separatrix location, source loading, coupling and geometry are all compatible explanations. Their contributions were not separated experimentally or computationally in this closed campaign. A claim that the cubic family cannot exhibit network memory would exceed the evidence.

Dependence on stimulus and regenerative mechanism

The history contrast disappeared when stronger stimulation caused the intact network to recruit already. It also disappeared at higher sampled gD and in the tested cubic formulation. This dependence is scientifically more informative than a claim that nonlinear networks simply bypass a bottleneck. The interface can limit a graded signal while a temporary reduction in loading changes whether regenerative recruitment occurs. The measured graded transmission fraction and the final recruitment fraction answer different questions.

The original motivation invoked maximum flow and Cheeger-type intuition. Maximum-flow/minimum-cut is a theorem about capacity-constrained flows, not a theorem exclusively about passive electrical circuits; Cheeger inequalities relate graph expansion and spectra. Neither directly establishes the transmission fraction in the present membrane equations. The results are therefore not counterexamples to those theorems. They concern the limits of transferring linear or static bottleneck intuition to nonlinear state selection.

Relation to bioelectric pattern memory

Experimental pattern-memory studies motivate this question, but a synthetic equilibrium difference cannot establish their molecular mechanism. BGR models already link local feedback to multicellular voltage patterns, and BEN models retain transient inputs. [Pietak and Levin, 2017](#); [Manicka and Levin, 2019](#). A recent geometric preprint also treats hidden regenerative state and temporal-profile dependence; its endpoint and evidence level differ from the present fixed-network assay. [Blattner, 2026, preprint](#).

The present model is useful because its final connectivity can be known exactly. Its biological applicability remains an open empirical question.

Limitations

First, the depolarizing mechanism is synthetic, the potassium kinetics are approximated, and no tissue-specific channel complement was identified. Second, the five configurations share a common origin and include only modest geometric changes. Third, the protocol anchors were selected using earlier behavior of the original network, and the cubic comparison was not matched for regenerative current strength or collective excitation threshold. Fourth, local stability and 5 s persistence are not a global proof of basin geometry or indefinite biological memory.

Fifth, the receiver definition and $\geq 50\%$ recruitment criterion are operational choices; no claim of biological optimality follows. Sixth, the main campaign included exploratory choices and later controls, transparently separated from the bounded final validation. Seventh, the repair comparison used a heuristic and a small fixed set of random orders. Finally, literature access limitations leave priority of the exact restoration experiment unresolved. These limitations affect interpretation without negating the reported computational contrasts.

Experimental/computational predictions

A falsifiable transfer prediction is that, in an independently characterized bistable preparation with controlled coupling, some fixed stimuli will produce different persistent electrical states after a temporary coupling reduction and full recovery, whereas stronger stimuli may erase the contrast by recruiting both histories. Testing this prediction would require measuring recovery of coupling as well as voltage persistence; removal of a blocker alone would not establish the matched-final-system condition.

A second prediction is conditional: local hysteresis will not reliably predict network history dependence unless the stimulation and coupling dynamics can access the alternative network state. These are interpretations and externally testable predictions of the completed study, not proposals to expand the present simulation campaign.

Conclusion

In the tested synthetic conductance-based network, transient perturbation of intercellular coupling altered stable state selection even after exact restoration of the original weighted connectivity. Local multistability provides a framework for interpreting this history dependence, while the negative cubic comparison shows that bistability and hysteresis alone were insufficient under the fixed stimulation conditions. The result is restricted to the specified synthetic models, networks and protocols.

Data and code availability

The accompanying reproducibility archive contains the frozen protocols, model code, simulation trajectories, analysis scripts and numerical validation records. The study's Zenodo identifier is [10.5281/zenodo.22802165](https://doi.org/10.5281/zenodo.22802165). Manuscripts and project documentation are available in the [GitHub repository](#).

Declarations

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AI-assisted preparation: OpenAI Codex was used to assist with computational implementation, calculations, figure preparation and manuscript editing.

Ethics statement: This computational study involved synthetic networks and no human participants or animal experiments.

Licensing: Original text, data and figures are licensed under CC BY 4.0; original code is licensed under the MIT License.

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Figure 1

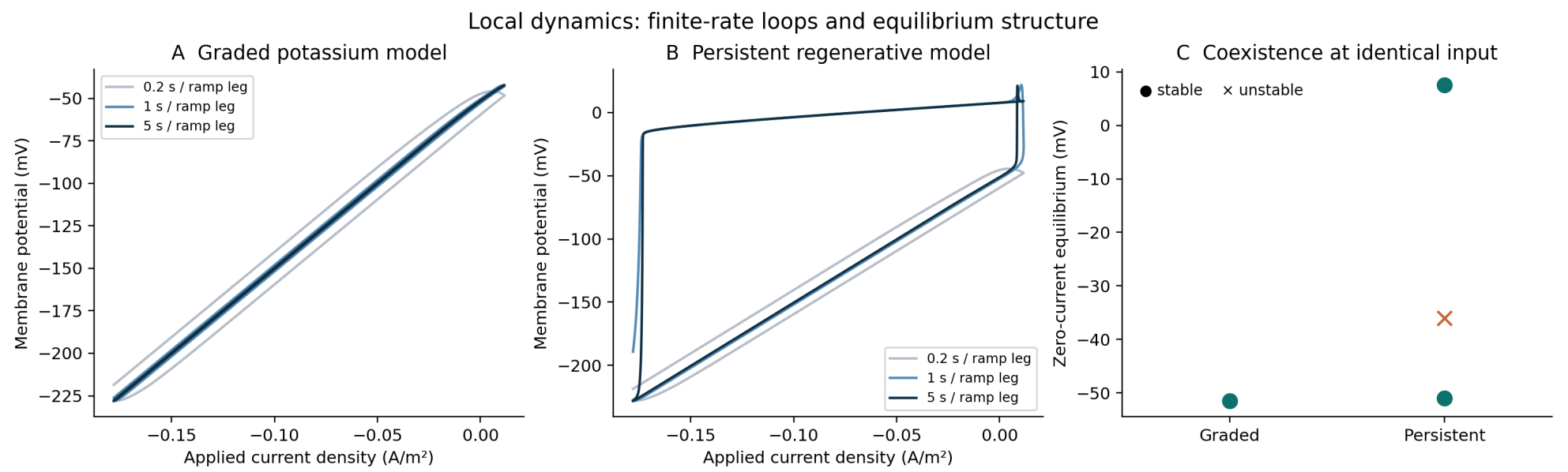


Figure 1. Local dynamics and equilibria. Finite-rate current ramps in the graded potassium and persistent models (0.2, 1 and 5 s per leg); zero-current stable equilibria are filled circles and the unstable equilibrium a cross. Voltages are mV, currents A/m^2 . The graded loop shrinks with slower ramps; the persistent 1 and 5 s ramps switch branches. The fast persistent ramp does not switch in the same way. The deliberately synthetic sweep extends below physiological voltages. Source: archived hysteresis trajectories and equilibrium records.

Figure 2

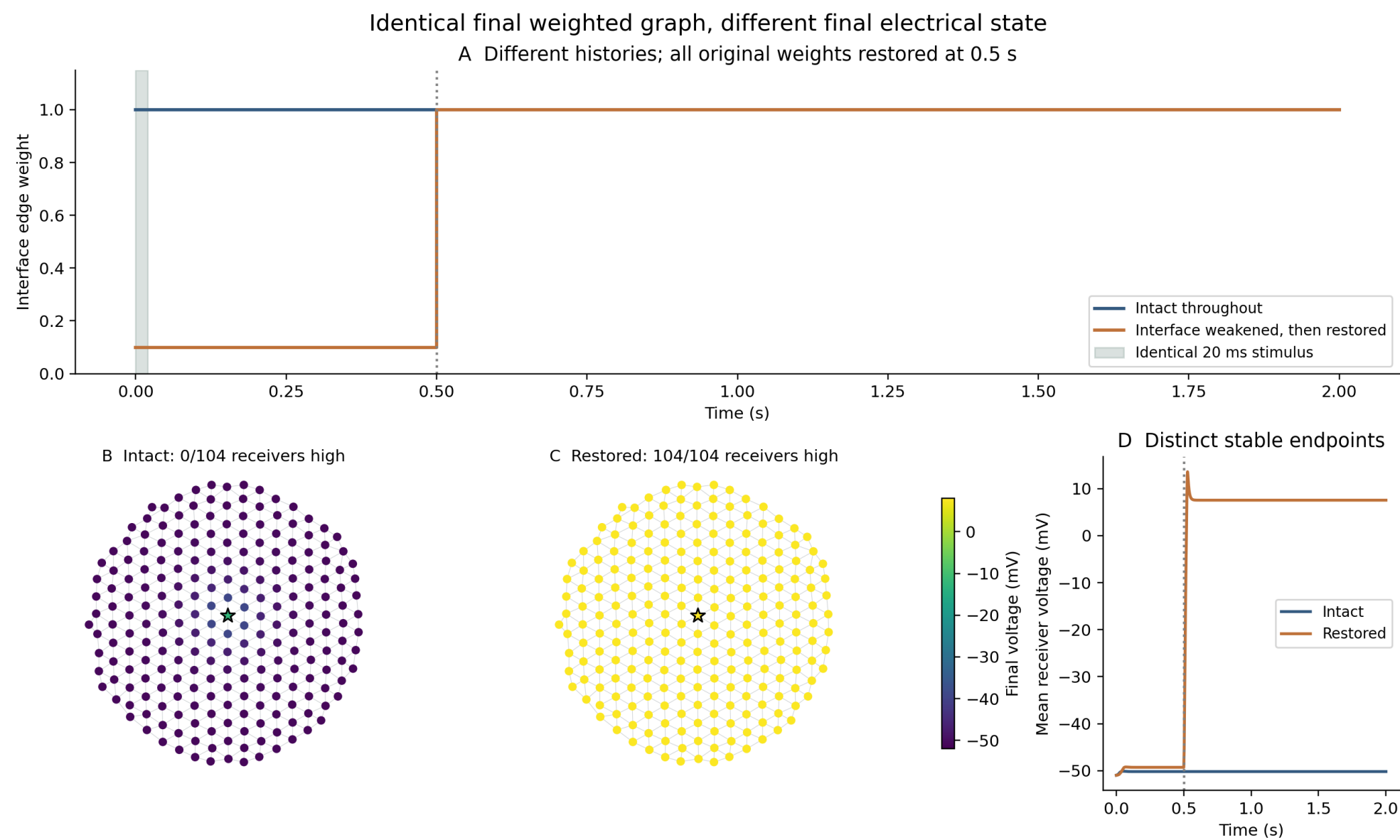


Figure 2. Matched final system after different coupling histories. A: interface weights, identical 20 ms pulse and exact restoration at 0.5 s. B-C: final voltage at 2 s in cases case0147 and case0159, with identical final weighted graphs; star marks the source. D: mean receiver voltage. The intact history has 0/104 high receivers and the restored history 104/104. Network Jacobians and separate 5 s perturbation-return checks support local stability. Voltage colors are shared between maps. No signal is asserted to cross an absent connection.

Figure 3

A — Geometry and source generalization | fixed 0.9× stimulus
All prespecified configurations; identical final connections within each pair

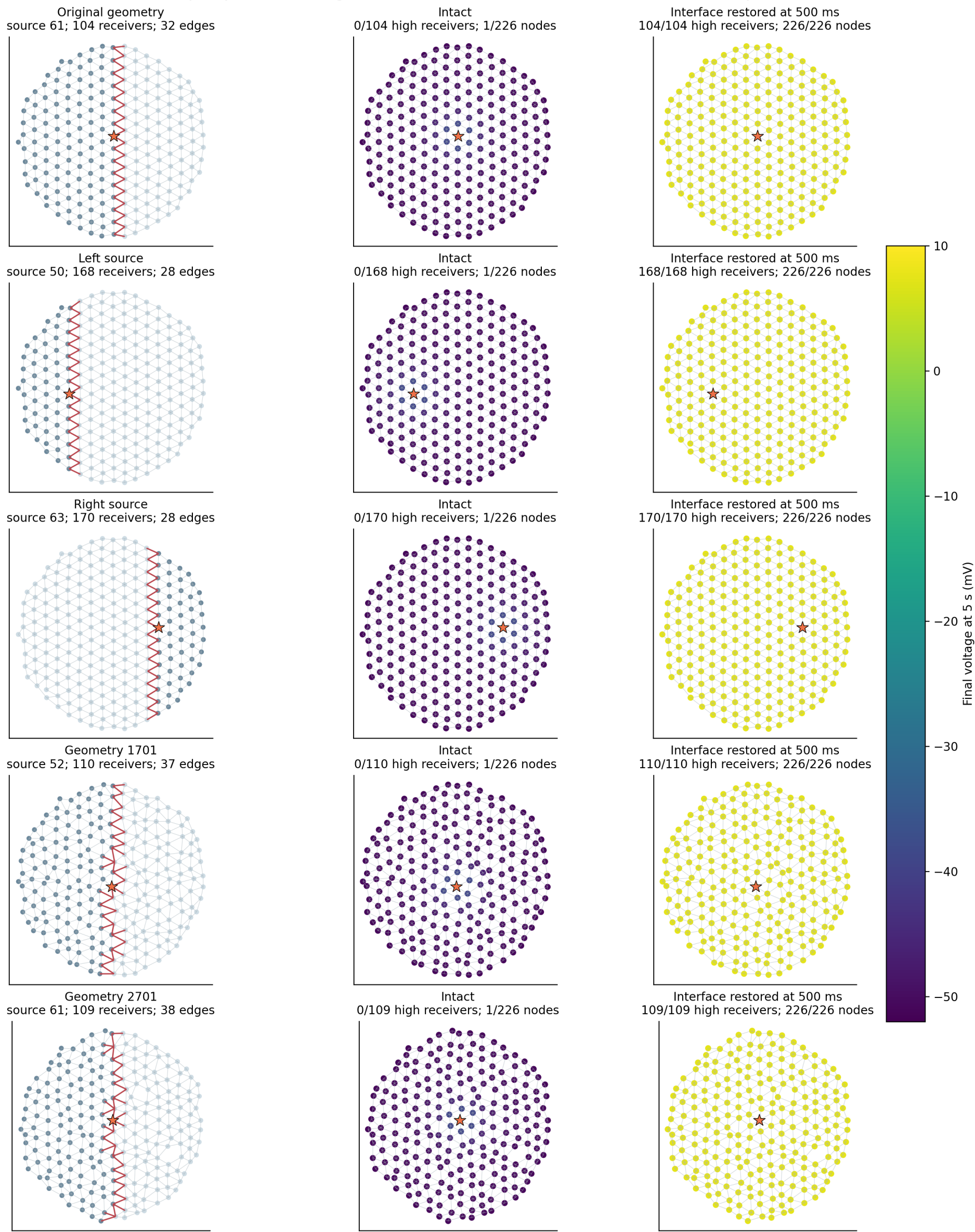


Figure 3. All five prespecified source/geometry configurations at the fixed 0.9 stimulus. Rows: historical, left source, right source, geometry 1701, geometry 2701. Columns: geometry/interface/source, intact endpoint, weakened interface followed by restoration at 500 ms. Endpoint colors show mV at 5 s. Source IDs are 61, 50, 63, 52 and 61; receiver counts are 104, 168, 170, 110 and 109. Every intact endpoint is localized and every restored endpoint collective. These correlated configurations do not establish generality across tissues.

Figure 4

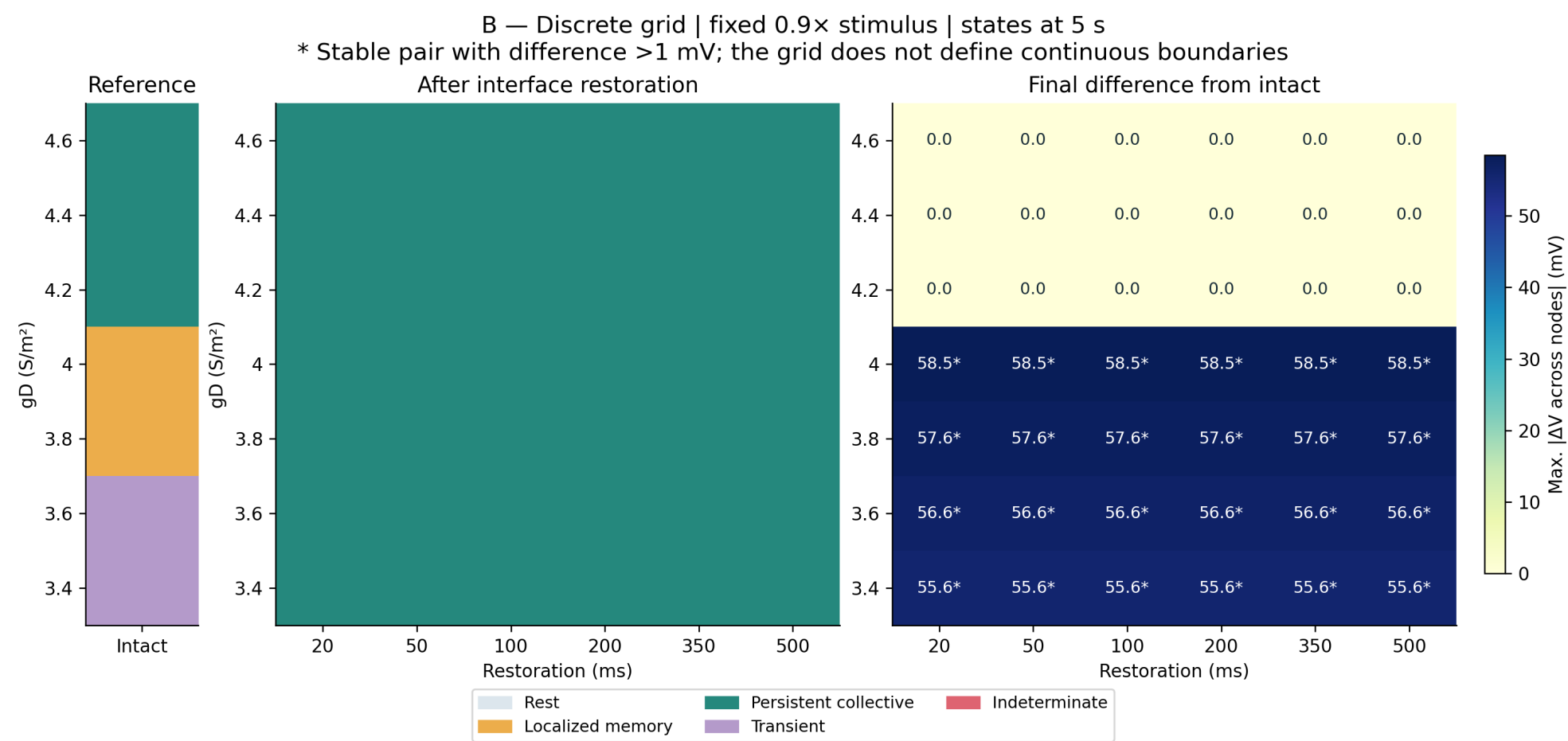


Figure 4. Discrete conductance-restoration grid at fixed factor 0.9. Left: intact reference; middle: restored interface; right: maximum absolute endpoint voltage difference across nodes, in mV. The grid covers $gD = 3.4\text{--}4.6\text{ S/m}^2$ and restoration times 20–500 ms. Asterisks mark stable matched-final-system pairs differing by more than 1 mV (24/42). Colors: pale blue, rest; orange, localized; purple, transient; green, collective; red, indeterminate. All restored cases are collective. No continuous boundary is inferred.

Figure 5

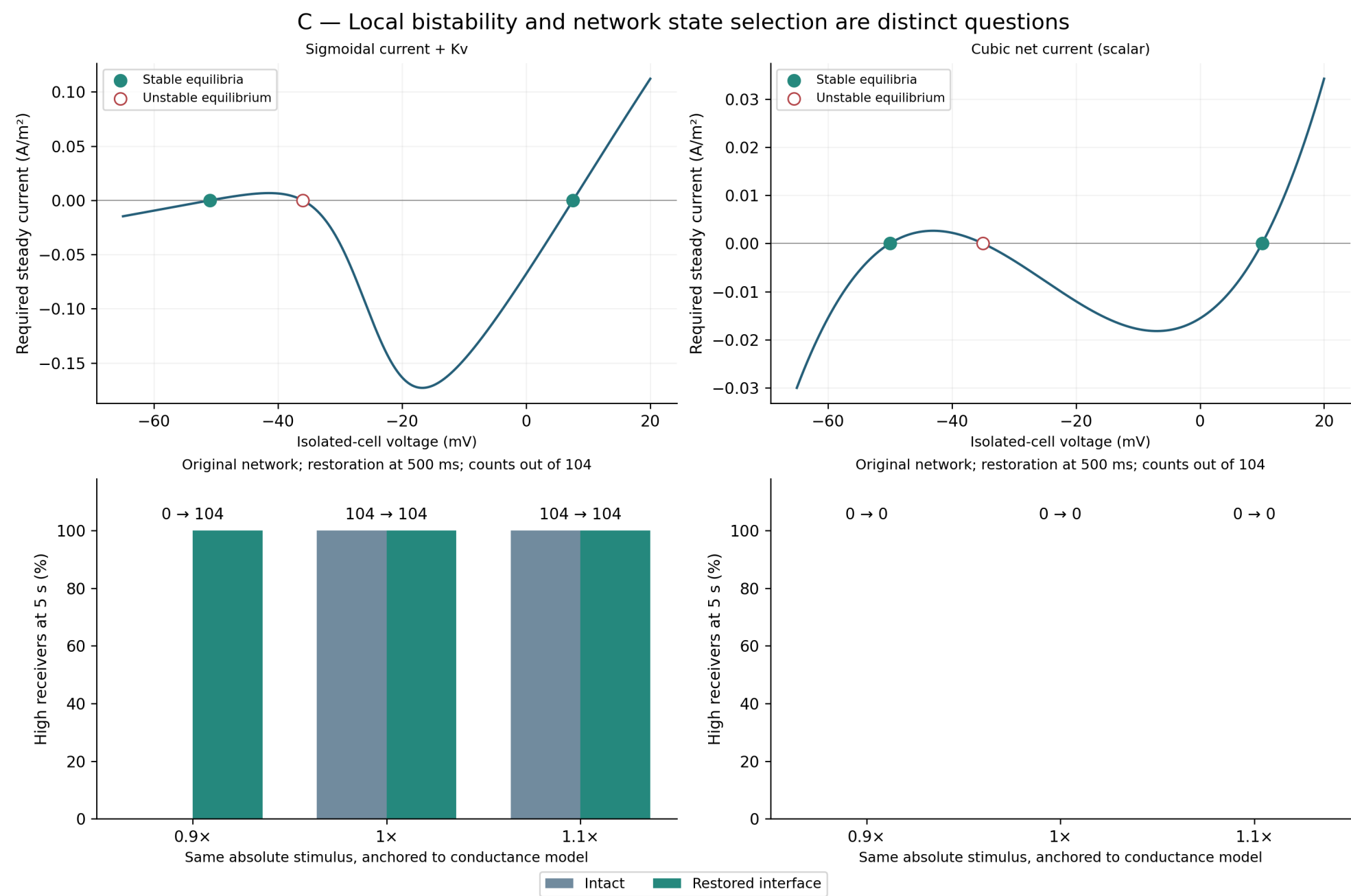


Figure 5. Local bistability and fixed-protocol network recruitment. Top: steady current required to hold an isolated voltage, using separate current scales for the conductance-based and cubic models; filled circles mark stable roots and open circles unstable roots. Bottom: high receivers at 5 s, intact versus restoration at 500 ms, at the same absolute amplitudes. Conductance-model recruitment changes from 0 to 104 receivers only at factor 0.9; cubic recruitment remains 0 to 0 at every factor. The cubic mechanism was not recalibrated after its negative result.

History-dependent state selection after transient coupling loss in a conductance-based bioelectric network

Supplementary material

Version 1.1 — 16 September 2026

This preprint has not been peer reviewed. Version 1.1 translates figure labels into English, updates author declarations and repository links, and expands the prior-literature discussion. Numerical results and conclusions are unchanged. Zenodo DOI: [10.5281/zenodo.22802165](https://doi.org/10.5281/zenodo.22802165).

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S1. Scope and equations

This supplement describes existing frozen computations. No new simulation or parameter search was performed for this revision. Existing figures were reexported with English labels using frozen data. The primary endpoints are synthetic electrical states, not anatomy, tissue regeneration or biological memory measurements. The graph has symmetric weights and an unnormalized Laplacian: $L(W) = \text{diag}(W \mathbf{1}) - W$. The coupling term conserves the summed network exchange current; membrane currents and applied stimuli need not conserve the summed voltage.

The conductance-based model is:

$$C_m \dot{V}_i = -G_j[L(W)V]_i - G_L(V_i - E_L) - G_K m_i(V_i - E_K) + g_D a_\infty(V_i) h_i(E_D - V_i) + I_i(t)$$

$$\dot{m}_i = [m_\infty(V_i) - m_i]/\tau_K, \quad m_\infty(V) = \text{logistic}[(V + 0.00076)/0.01409]$$

$$a_\infty(V) = \text{logistic}[(V + 0.025)/0.004].$$

$$\dot{h}_i = [h_\infty(V_i) - h_i]/0.1, \quad h_\infty(V) = \text{logistic}[-(V + 0.040)/0.004].$$

Here $\text{logistic}(x) = 1/(1 + \exp(-x))$. For the persistent variant $h = 1$; for the graded variant $g_D = 0$. The excitable variant uses the inactivation equation above with a 0.1 s time constant. These are independent reduced ODE simulations, not new native BETSE executions.

The cubic model replaces the full membrane-current law and has no potassium gate:

$$\dot{V}_i = \frac{(V_i - V_L)(V_i - V_U)(V_H - V_i)}{\tau_c(V_H - V_L)^2} - \frac{G_j}{C_m}[L(W)V]_i + \frac{I_i(t)}{C_m}$$

S2. Parameters, units and stimulus anchors

Quantity	Value	Meaning
Cm	0.01 F/m ²	Effective membrane capacitance density
GL; GK	1; 2 S/m ²	Leak; potassium conductance density
EL; EK; ED	−0.05; −0.07967507185138671; +0.05 V	Fixed reversal potentials
τK; τh	0.0148; 0.1 s	Constant effective kinetics
Gj	1 S/m ² in central comparison	Common normalization for coupling
gD	4 S/m ² nominal; 3.4, 3.6, 3.8, 4, 4.2, 4.4, 4.6 in validation	Synthetic regenerative conductance
Cubic VL; VU; VH	−0.05; −0.035; +0.01 V	Two stable roots separated by an unstable root
Cubic τc	0.003125 s	Low-state slope approximately −80/s
Persistent anchor	0.082373046875 A/m ²	Earlier original-network upper threshold bracket
Persistent lower bracket	0.08232421875 A/m ²	Earlier development; not independent validation
Excitable anchor	0.12783203125 A/m ²	Separate earlier network anchor
Amplitude factors	0.9, 1.0, 1.1	No validation geometry/cubic recalibration
Pulse	20 ms in regenerative network tests	Applied to one source
Restoration grid	20, 50, 100, 200, 350, 500 ms	Original-geometry validation map
Final validation horizon	5 s; samples every 1 ms	Not proof of unlimited persistence

The potassium midpoint (−0.76 mV) and slope (14.09 mV) derive from Dong et al. (2022), DOI 10.3389/fphar.2022.965086. Their heterologous hKv1.5 measurements do not identify the whole synthetic membrane. The constant τK approximates one reported deactivation timescale. Authenticated Channelpedia data are neither bundled nor used to identify the added regenerative current.

Main temporal tests comprise equal-charge (0.0001 C/m²) and equal-current (0.02 A/m²) series at 1, 10 and 100 ms, τK of 14.8 or 59.2 ms and Gj of 1 or 4 S/m². Exact per-case values are in `protocolo.json`; these graded protocols differ from the later regenerative recruitment assay.

S3. Frozen protocols and enumeration

The main protocol was frozen locally on 14 September 2026 at 21:09:27.908685 UTC, after isolated-cell and original-network development. It contains 487 labels: 144 temporal, 144 memory, 60 conductance-sensitivity, 126 repair, nine no-stimulus and four never-restored conditions. Subsequent controls and corrections remain separately documented. The main freeze was not external preregistration.

Validation was frozen on 15 September 2026 at 13:58:30.658348 UTC. Its 136 unique network cases contribute 65, 49 and 24 block memberships; two cases are shared. Six isolated protocols bring the scientific-condition count to 142. Network numerical verification comprises 136 primary, 136 refined and 44 Radau runs; isolated verification adds 18 runs. These technical runs are not biological replicates.

Original main protocol SHA-256:

5cbb0be57d3d0a06bf0f8ff971c9db2ded15500a959596ed44f1eb7db40680ba.

Original validation protocol SHA-256:

a9d5d8078bfef94c02864a6c7753946882259e0846e44b8541cadb715deac4fb.

The public validation copy changes only its private executable path. ZENODO_PROVENANCE.md and SOURCE_PROVENANCE.json record its distinct public hash and all dependent manifest changes; the original freeze is not attributed to altered bytes.

S4. Configurations and matched histories

Configuration	Source ID	Receivers	Interface edges
historica	61	104	32
fonte_esquerda	50	168	28
fonte_direita	63	170	28
geometria_1701	52	110	37
geometria_2701	61	109	38

All five configurations contain 226 nodes and 622 edges. The two geometry variants use fixed jitter seeds 1701 and 2701 with standard deviation 0.1 times the original median edge length, followed by a Delaunay/minimum-spanning-tree reconstruction retaining 622 edges. Their undirected symmetric edge differences are 12 and 10: six and five replacements, not 12 and 10 replacements. Source variants use the specified coordinate-quantile rule; receiver masks are fixed before outcomes.

Every interface weight is reduced from 1 to 0.1 until restoration. Original-interface loss is $32 \times 0.9 = 28.8$ weight units. Random controls match edge count and per-edge loss within each configuration. Exact restoration reinstates every original weight without resetting voltage or gates. Once the pulse ends, paired histories have identical final W , membrane equations, parameters and zero external input. Different trajectories can therefore reach distinct states under the same final vector field.

S5. Hysteresis and stability checks

The graded model has one zero-current equilibrium at -51.497483 mV with largest real eigenvalue $-86.441007/s$. Its loop areas at 0.2, 1 and 5 s per leg are 0.0032821472, 0.0006794626 and 0.0001363633 V A/m². Slower-loop shrinkage and convergence to a common rest support kinetic lag.

Persistent-model zero-current roots are -50.971228 , -36.016572 and $+7.547151$ mV, with largest real eigenvalues -79.219846 , $+337.710679$ and $-104.270068/s$. The 1 and 5 s loop areas are 0.0247225890 and 0.0238513057 V A/m², with switching in both directions. The 0.2 s ramp does not switch equivalently. Loop area alone is not proof of quasistatic hysteresis or network memory; the current sweep deliberately reaches nonphysiological voltages.

The intact and restored spatial witnesses, case0147 and case0159, have full-network largest real eigenvalues -59.816278 and $-80.906415/s$. They retain 1 versus 226 high nodes and 0 versus 104 high receivers. Recorded residuals are 1.15×10^{-12} and 1.81×10^{-8} V/s; separate small-perturbation checks returned to these states. These checks support local stability and do not map global attraction basins.

S6. Repair and excitable-model results

Repair restores 4, 8 or 16 interface edges before stimulation. The directed ranking uses the integrated absolute voltage difference across edges during an earlier graded response; three fixed random orders are comparators. In all 27 graded comparisons, the directed peak exceeds the random peak by 0.00810–0.04001 mV. At persistent-model factor 0.9, directed repairs leave 0/104 receivers high whereas the sampled random repairs recruit 104/104. At higher tested amplitudes both strategies are collective. This is a heuristic endpoint-dependent ranking, not an optimal repair theorem (Figure S3).

The inactivating extension shows transient excitability in isolated tests and returns to rest. A 2 ms suprathreshold isolated pulse produces a delayed large excursion; two-pulse tests distinguish short-interval suppression from recovery. In the original network, restoring the interface at 20 or 50 ms permits large receiver excursions, whereas restoration at 200 or 500 ms does not in the corresponding weak-interface comparisons. Some other conditions have small threshold crossings without large regenerative excursions. The bottom row of Figure S1 counts threshold crossings and must not be interpreted as a spike count. No persistent attractor memory is inferred for the excitable variant.

S7. Random, never-restored and ablation controls

At the lowest fixed stimulus, random weakening recruits collectively in four of the five validation configurations; the right-source configuration remains localized. The central response is therefore not exclusive to the chosen interface.

Five persistent never-restored interface controls end with 122, 58, 56, 226 and 117 high nodes, respectively. Their high-receiver counts are 0, 0, 0, 110 and 0. One geometry thus recruits without restoration, limiting any claim that restoration is necessary for propagation. Never-restored conditions cannot themselves establish history dependence under identical final connectivity. A completely cut interface cannot transmit across the missing path; recruitment after reconnection is a different event.

Validation includes six no-stimulus cases and twelve additional-current ablations. No-stimulus departure is at most 6.94×10^{-18} V. Ablations return to rest, although some trajectories cross the operational threshold transiently. The cubic ablation leaves pure leak, whereas conductance-model ablation removes the additional depolarizing current and retains leak plus potassium dynamics.

S8. Negative cubic result and limits of transfer

The cubic model has roots -50 , -35 and $+10$ mV with local eigenvalues -80 , $+60$ and -240 /s. Four perturbed zero-current holds and two ramps (1 and 5 s legs) verify bistability and hysteretic switching before the network tests. Despite source activation, all 21 stimulated network conditions return to rest without receiver recruitment. All twelve paired intact/restored comparisons have no history-dependent endpoint contrast; their maximum voltage difference is 6.94×10^{-18} V.

Bistability and hysteresis alone were insufficient to produce network-level history dependence under the fixed stimulation protocol. The comparison did not match the regenerative current shape, collective threshold or basin accessibility. No cubic retuning was performed. This is a protocol-specific negative result, not proof that scalar cubic networks can never exhibit history dependence.

S9. Numerical verification and effective outputs

The historical main verification initially failed six cases at a 0.02 mV criterion, with maximum discrepancy 0.0317561 mV. Their original files remain. Separately labeled replacements in `precisao_corrigida` reduced the maximum refined comparison to 0.000210172 mV; Radau checks retained the classifications. Effective results are `metricas_conferidas.csv` and `resultados_conferidos.jsonl`. The historical failed `verificacao.json` must not be mistaken for the final correction status.

All 136 validation conditions passed refinement without replacement. Maximum primary/refined difference was 0.000214893 mV; maximum Radau/refined difference was 1.115049×10^{-11} V in 44 prespecified comparisons. Effective validation trajectories are in `refinadas`, linked by `effective_source` in `resultados_conferidos_validacao.jsonl`. All accepted validation endpoints satisfy max voltage derivative $<10^{-5}$ V/s, gate derivative $<10^{-4}$ /s where applicable, and full-Jacobian maximum real eigenvalue $<-10^{-6}$ /s. History contrast additionally requires >1 mV paired endpoint difference. Recruitment uses the model-specific intermediate root and a $\geq 50\%$ receiver threshold; it is not an action-potential definition.

Primary DOP853: `rtol` 10^{-10} , `atol` 10^{-12} , max step 2.5 ms. Refined DOP853: 10^{-12} , 10^{-14} , 1 ms. Radau: 10^{-11} , 10^{-13} , 2.5 ms. Event boundaries are explicitly segmented. The completed validation audit rechecked all 136 trajectories and 87 paired comparisons. During deposit preparation, read-only recomputation again agreed with stored receiver counts and peaks; all 959 source scientific files retained their recorded hashes.

S10. Reproduction and provenance

The ZIP preserves both frozen scientific trees and their two required inputs. Full trajectories are retained to permit independent endpoint, transient and solver audits, including the six failed original main cases; compressed arrays are stored once rather than duplicated in display-oriented folders. Exploratory calibration records explain the origin of the fixed anchors and are historical evidence, not new validation. Private literature caches, authenticated datasets, environments, Git history and journal-planning files are excluded.

Original scientific environment: Python 3.9.6, NumPy 1.26.4, SciPy 1.13.1, Matplotlib 3.7.5. Dependencies are specified, not redistributed. Use `verify_bundle.py` after extraction for byte-level checks without model execution. Reproduction scripts prepare new sibling directories by default and run the archived protocol only with the explicit `--run` option. Main reproduction reruns the 487-case experiment and numerical correction, not every historical exploratory or supplementary procedure; final validation reproduction includes the 136 network and six isolated protocols. Figure-only scripts and the full scientific scripts are provided separately. Consult `README_ZENODO.md` for exact commands and scope.

Figure S1

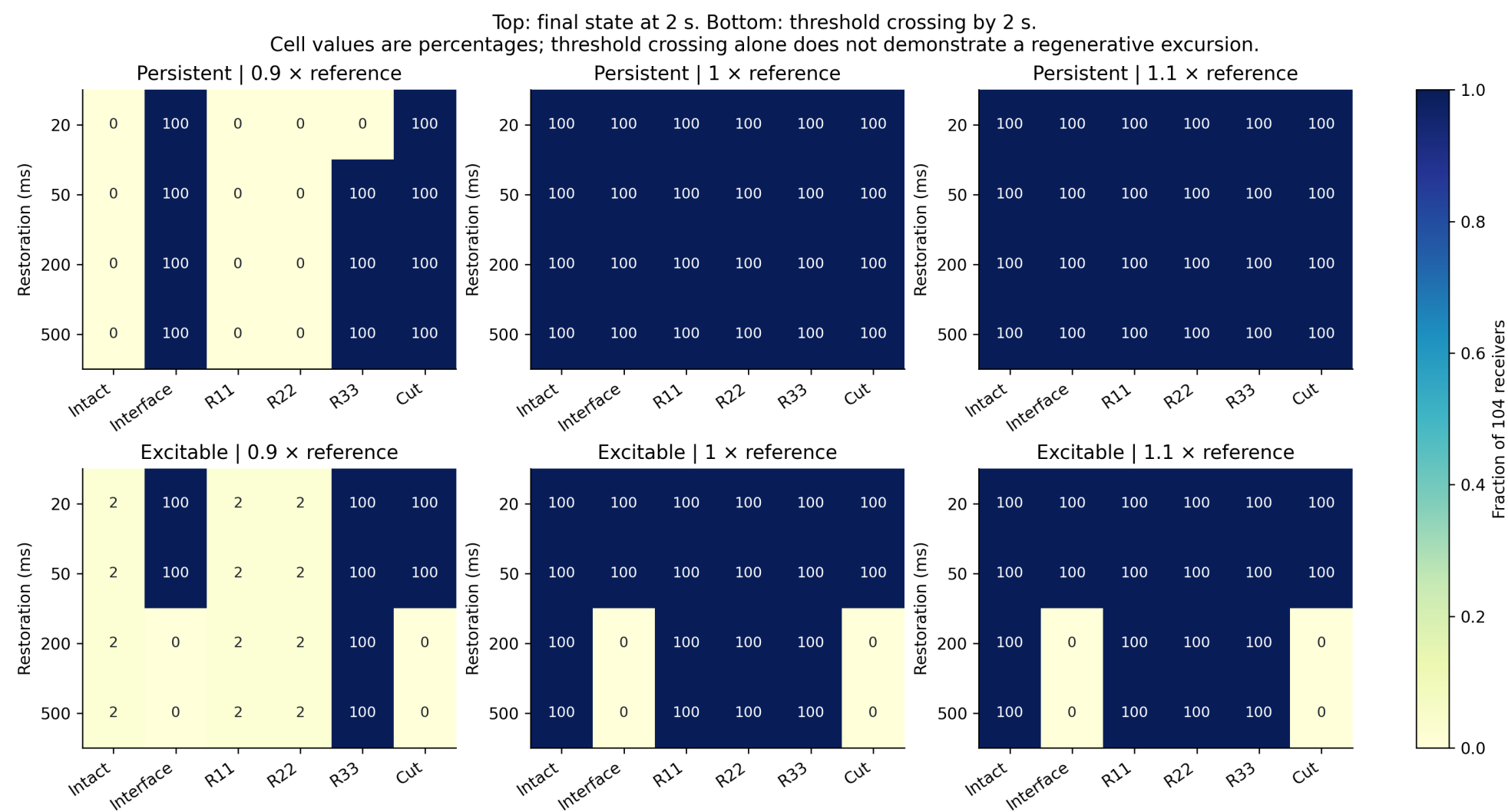


Figure S1. Main-campaign recruitment by restoration time and amplitude. Top: persistent-model final high-receiver fraction at 2 s; bottom: excitable-model fraction ever crossing the fixed threshold by 2 s. Color scale is a fraction, cell annotations are rounded percentages. Crossing that threshold alone is not evidence of a regenerative excursion. Intact = original network; Interface = 10%-weight interface; R11/R22/R33 = sampled random weakenings; Cut = complete cut followed by restoration. Original source/receiver masks are fixed.

Figure S2

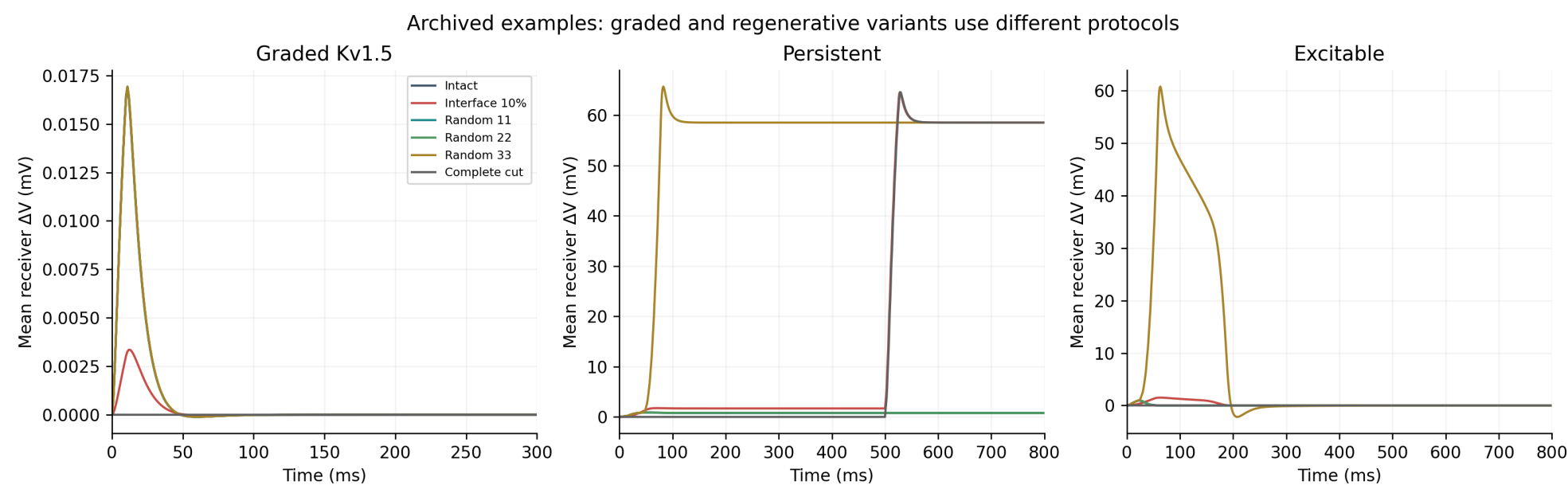


Figure S2. Archived example receiver responses for graded, persistent and excitable variants. Axes: time (ms), mean receiver voltage change relative to rest (mV). Protocols differ between the graded and regenerative examples, as specified in the original figure script; panel heights must not be interpreted as a dose-matched effect size. Interface 10% denotes nonzero coupling; Complete cut denotes a complete cut followed by scheduled restoration.

Figure S3

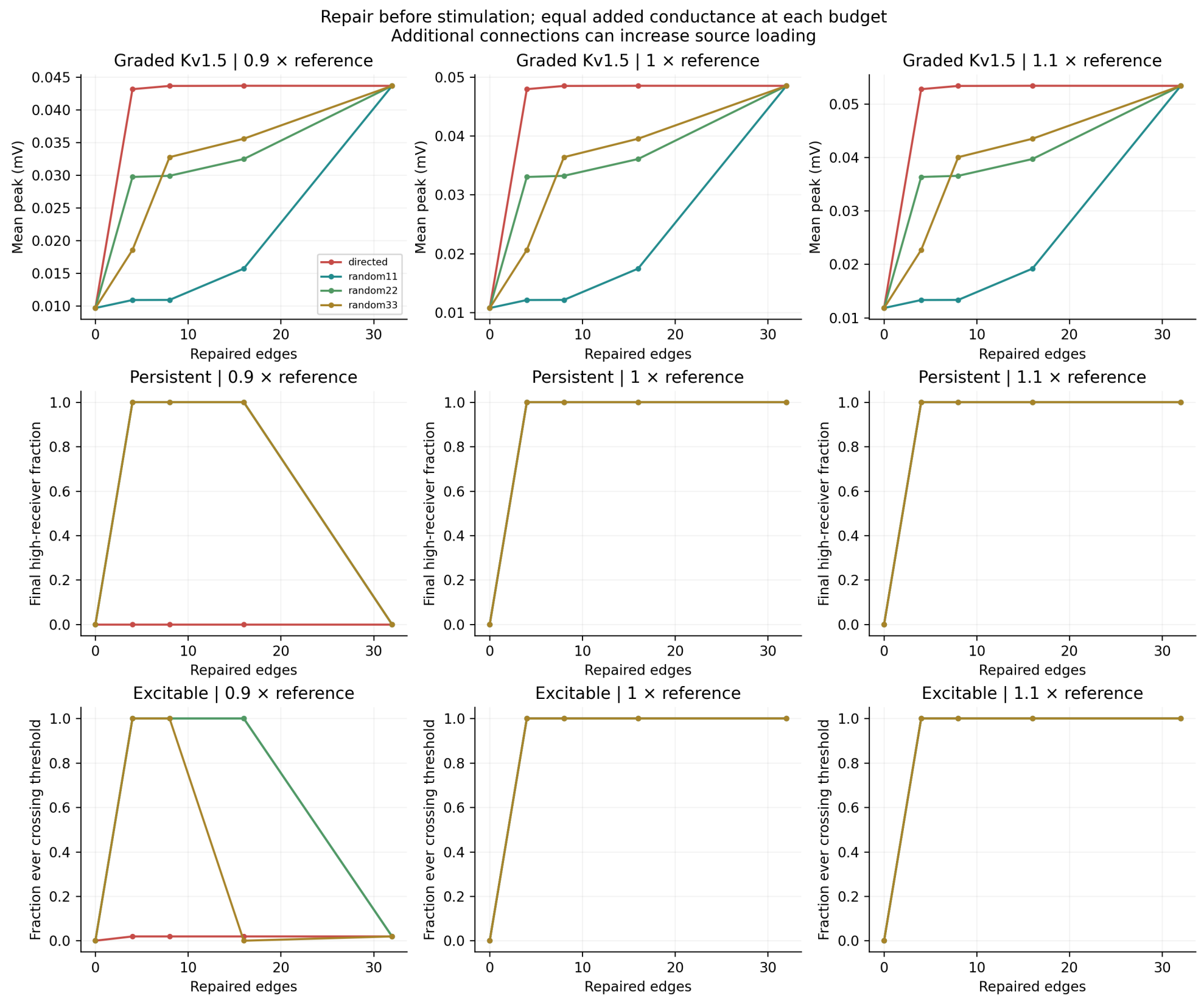


Figure S3. Partial repair before stimulation, a separate assay from complete post-perturbation restoration. Columns: amplitude factors 0.9, 1 and 1.1. Rows: graded peak (mV), persistent final fraction, excitable ever-crossing fraction. Horizontal axis: repaired interface edges; 4, 8 and 16 are tested partial budgets and endpoints show damaged/intact references. Directed repair is compared with three fixed random orders. At factor 0.9, the graded ranking does not preserve persistent recruitment ordering. Line segments guide the eye and do not supply untested budgets.